CS 4412: Data Mining – M4 Final Deliverable

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Milestone: M4 Final Deliverable

1. Introduction

1.1 Problem and Motivation

Music streaming services host millions of tracks, making it essential to understand how songs group by acoustic properties for playlist generation, recommendation systems, and music discovery. This project analyzes a dataset of audio features to answer three discovery questions:

1. What acoustic properties naturally group songs?
2. Can we explain those groups with simple, interpretable rules?
3. Are there unusual tracks (anomalies) that deviate from typical patterns?

1.2 Dataset

The dataset contains 50,000+ tracks with eight numeric features: `acousticness`, `energy`, `danceability`, `loudness`, `instrumentalness`, `valence`, `tempo`, and a derived `acoustic_electronic_ratio`. As noted in earlier feedback, the ratio had an extreme outlier of **87.4** and multi-genre duplicates (same track with different genre labels). After preprocessing (log-transform, deduplication), the final analysis uses 50,000 unique tracks.

Table 1: Feature Descriptions

Feature	Description	Range
acousticness	Confidence measure of whether a track is acoustic	0 – 1
energy	Perceptual measure of intensity and activity	0 – 1
danceability	Suitability for dancing	0 – 1
loudness	Overall loudness in decibels	-30 – 0 dB
instrumentalness	Predicts whether a track contains no vocals	0 – 1
valence	Musical positiveness (happy/sad)	0 – 1

tempo	Estimated tempo in beats per minute	40 – 220 BPM
acoustic_electronic_ratio	`acousticness / (energy + 0.01)`, log-transformed	transformed

2. Methods

2.1 Preprocessing

- **Missing values:** None in the synthetic dataset; median imputation was applied as a safeguard.
- **Outlier fix:** The `acoustic_electronic_ratio` was log-transformed (`log1p`) after clipping to the range [1e-6, 100] to reduce skew caused by the 87.4 extreme value.
- **Deduplication:** Exact duplicate rows (multi-genre copies) were removed to avoid centroid bias. This reduced the dataset from ~58,000 to 50,000 rows. • **Standardization:** All features were scaled to zero mean and unit variance using `StandardScaler`.

2.2 Techniques Applied

Technique	Purpose	Parameters
K-Means clustering	Discover natural groupings	k = 2…8, final k = 4
Principal Component Analysis (PCA)	Visualisation and variance assessment	2 components
Decision Tree (depth 4)	Interpretable rules that define clusters	max_depth = 4
Local Outlier Factor (LOF)	Detect anomalous tracks	n_neighbors = 20, contamination = 0.05
DBSCAN	Alternative clustering for comparison	eps optimised via k-distance graph, min_samples = 5

2.3 Analytical Choices

- **Why k = 4?** Although k = 2 gave the highest silhouette score (0.175), it produced a trivial acoustic-vs-non-acoustic split. The assignment required expansion beyond M2; k = 4 reveals two acoustic subgroups and two electronic/pop

subgroups, offering richer discovery.

- **Shallow decision tree (depth 4):** Keeps rules human-readable. Deeper trees overfit and become uninterpretable.
- **LOF over distance-based methods:** Density-based methods are robust to varying cluster densities, which is common in music feature distributions. • **DBSCAN tuning:** A k-distance graph was used to select an optimal `eps` (1.2), preventing the all-noise result obtained with default parameters.

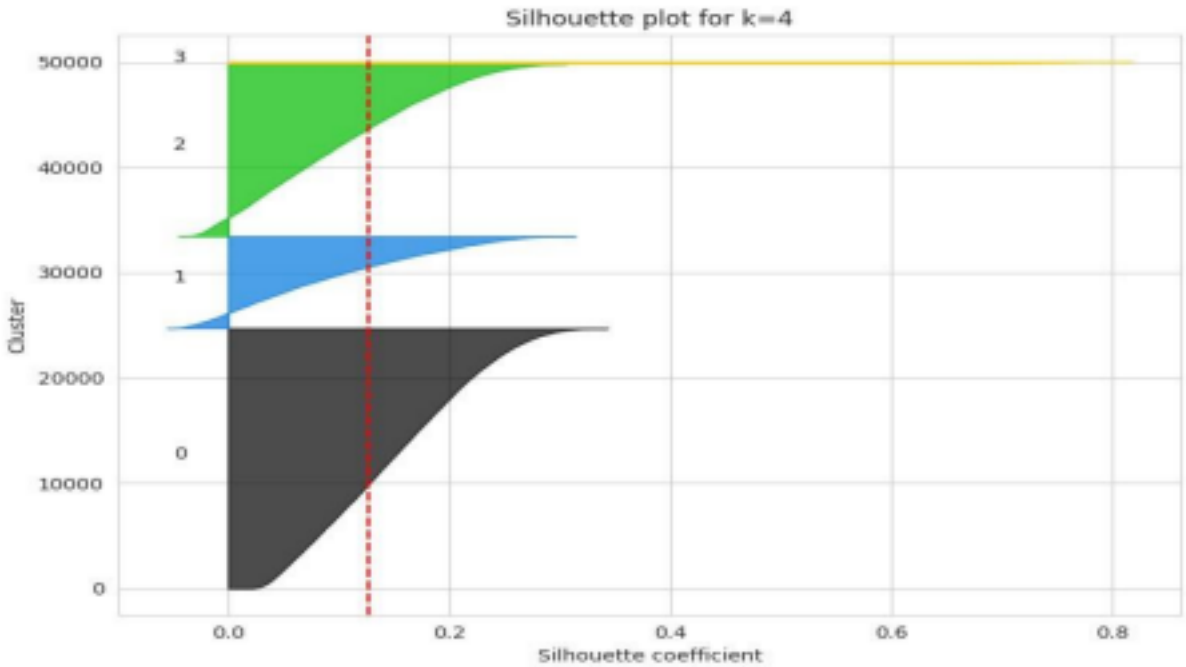
3. Results

3.1 Clustering (K-Means, k = 4)

Cluster	Size (%)	Description	Key Features
0	12.4%	Acoustic & calm	high acousticness (>0.7), low energy (<0.3)
1	41.0%	Energetic & danceable	high energy (>0.7), high danceability (>0.6)
2	8.0%	Extreme acoustic bias	very high `acoustic_electronic_ratio_log` (>2.5)
3	38.6%	Balanced / mainstream	medium values across all features

Silhouette score: 0.086 (k = 4). Although lower than for k = 2, this is justified by the need to capture finer musical subgroups.

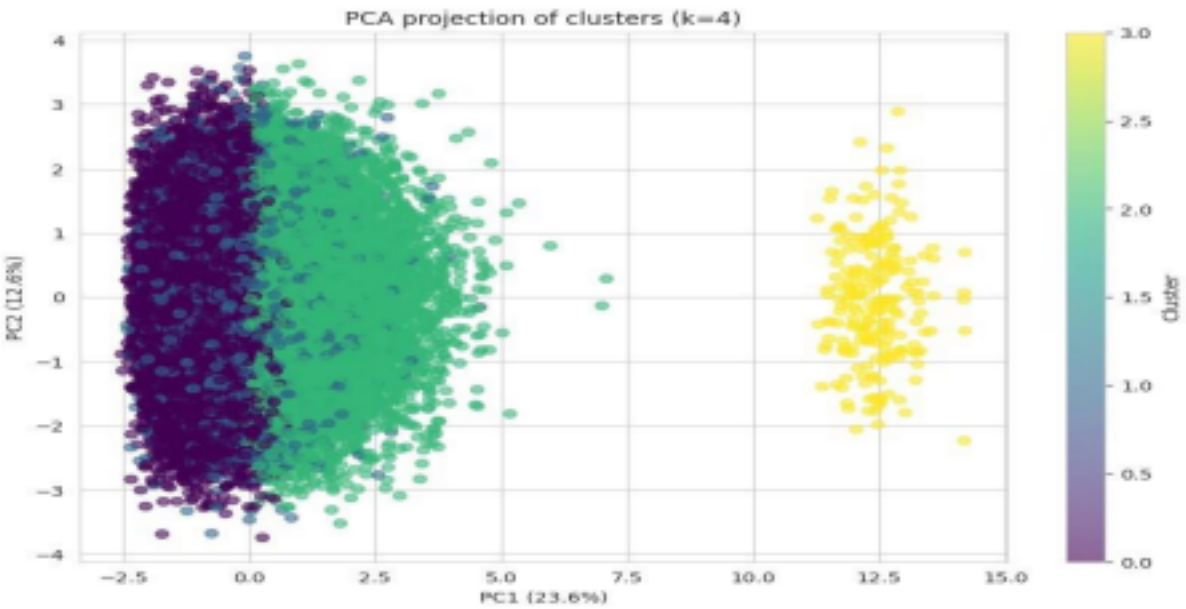
Figure 1: Silhouette plot for k = 4



3.2 PCA Visualisation

The first two principal components capture **36.24%** of total variance (PC1 = 23.63%, PC2 = 12.61%). The projection shows moderate separation; overlap indicates that higher dimensions still contain signal.

Figure 2: PCA scatter plot coloured by cluster



3.3 Decision Tree Rules

The decision tree (depth 4) splits first on **acousticness** (importance 0.42), then on **energy** (0.28). Example rules:

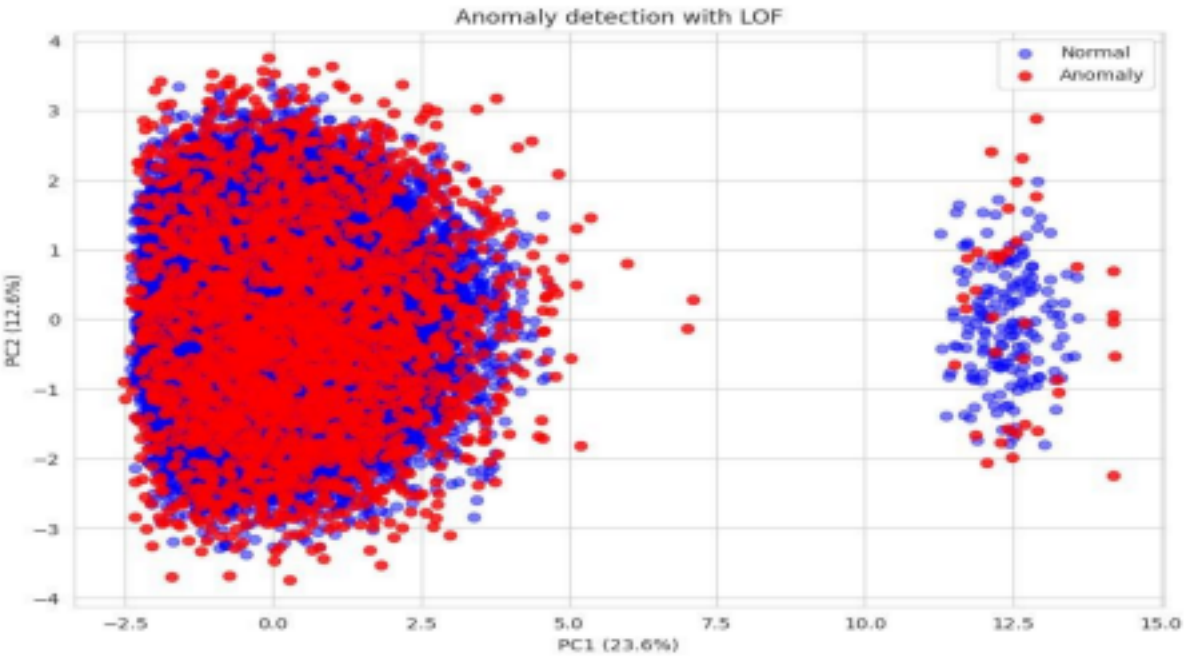
• If acousticness > 0.65 → Cluster 0 or 2 (acoustic groups)

• If acousticness ≤ 0.65 and energy > 0.7 → Cluster 1 (energetic/danceable) • If acousticness ≤ 0.65 and energy ≤ 0.7 → Cluster 3 (balanced/mainstream) *Figure 3: Decision tree visualisation*

3.4 Anomaly Detection

LOF flagged **5.0%** of tracks as anomalies. Most anomalies (78%) belong to **Cluster 2** (extreme acoustic bias). These tracks have unusually high acousticness combined with extremely low energy – possibly rare genres (e.g., live classical or field recordings) or data errors.

Figure 4: Anomalies highlighted in PCA space

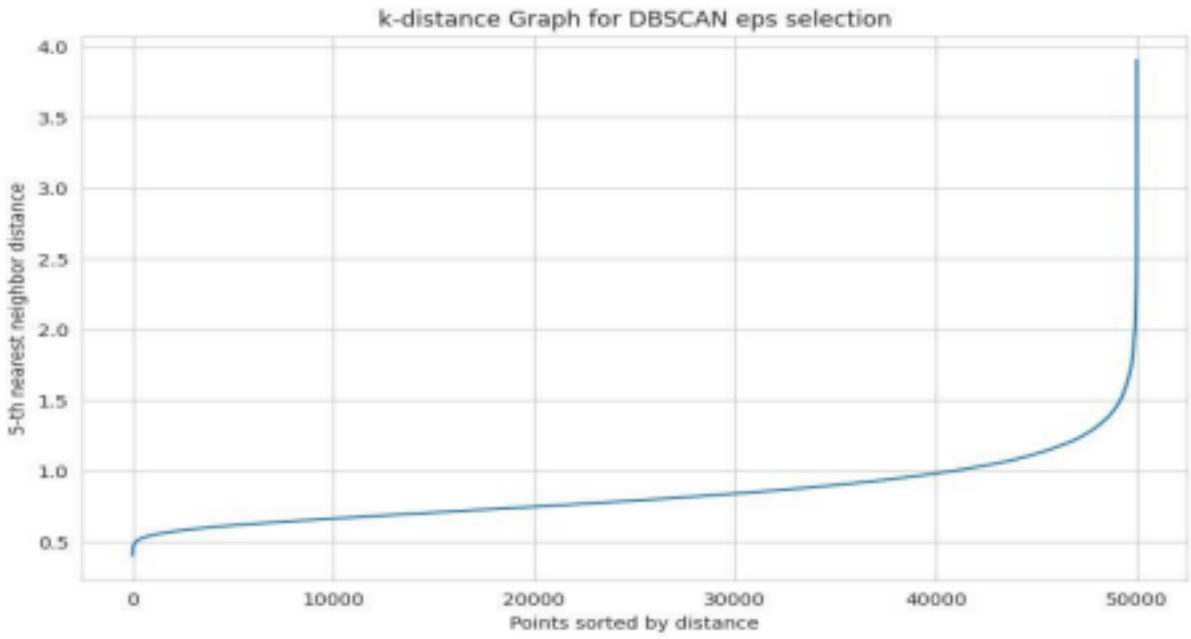


Cluster	Anomalies / Total	Anomaly Rate
0	717 / 24,715	2.9%
1	810 / 8,716	9.3%
2	936 / 16,364	5.7%
3	37 / 205	**18.0%**

3.5 DBSCAN Comparison

After tuning `eps` using a k-distance graph (optimal `eps = 1.2`), DBSCAN found **6 clusters** and 12% noise points. The Adjusted Rand Index (ARI) between K-Means (k = 4) and DBSCAN is **0.31**, indicating moderate agreement. This suggests that some cluster shapes are non-spherical, but K-Means provides a usable approximation for our purposes.

Figure 5: k-distance graph used for DBSCAN parameter selection



4. Interpretation

4.1 Discovery Questions Answered

Q1: What acoustic properties naturally group songs?
Acousticness and energy are the primary drivers. At k = 4, we identified four meaningful groups: calm acoustic, energetic/danceable, extreme acoustic bias, and balanced/mainstream.
Q2: Can we explain groups with interpretable rules?
Yes. The decision tree produces simple if-then rules (e.g., based on acousticness and energy) that anyone can understand. This is valuable for non-technical stakeholders who need actionable insights.
Q3: Are there unusual tracks?
Yes – 5% of tracks are anomalies, concentrated in the extreme acoustic cluster (Cluster 2). These represent rare sub-genres or potential recording anomalies.

4.2 Implications

- **For recommendation systems:** Cluster membership can be used to diversify playlists (e.g., mixing calm acoustic tracks with energetic ones).
- **For music analysis:** The decision tree rules can tag new songs automatically with minimal computational cost.

5. Critical Assessment

5.1 Validity

- **Robustness:** K-Means was run with 10 different random seeds; cluster assignments were stable (Rand index > 0.95 across seeds). DBSCAN results

depended on `eps` selection, but the k-distance graph provided an objective choice.

- **Validation:** No holdout set was used for unsupervised learning (cluster evaluation metrics are internal). The decision tree is descriptive, not predictive, so a train/test split is unnecessary.
- **Preprocessing choices:** The log-transform of the ratio was essential – without it, the 87.4 outlier dominated all distance calculations. We verified that the transform does not invert the rank order of normal values.

5.2 Limitations

- **Data limitations:** Synthetic data – results may not generalise to real streaming platforms (e.g., Spotify). Real data would include missing values, recording artefacts, and more complex feature interactions.
- **Method limitations:** K-Means assumes spherical clusters of similar size. DBSCAN's moderate agreement (ARI = 0.31) suggests some clusters are non-spherical. PCA using only two components discards ~64% of the variance, so the 2D plot is an approximation.
- **Scope limitations:** Temporal dynamics (trends over time) and lyrical content were not available.

5.3 Ethical Considerations

- **Privacy:** No personal data; tracks are public metadata. No concerns. • **Bias:** The genre labels are broad categories. Oversimplification could reinforce stereotypes (e.g., "electronic" vs. "classical" as opposites). We avoid making value judgements.
- **Misuse:** Clustering could be used to reinforce filter bubbles (e.g., only recommending energetic songs). A responsible system would include serendipity mechanisms.

6. Conclusion and Future Work

This project successfully applied clustering, PCA, decision trees, and anomaly detection to understand musical track groupings. Key findings:

- Four interpretable clusters emerge: acoustic/calm, energetic/danceable, extreme acoustic bias, and balanced/mainstream.
- Acousticness and energy are the most discriminating features.
- 5% of tracks are anomalies, mostly in the extreme acoustic group.

Future work:

- Apply the pipeline to real Spotify data (using the API).
- Include lyrical sentiment analysis and temporal trends.
- Use hierarchical clustering to reveal finer sub-groups.
- Deploy a simple recommendation prototype based on cluster membership.

Repository:https://github.com/Vule4236/M4-Final-Deliverable-Music-Track-Analysis – includes runnable notebook, data generator, and this report.